Lab 06: Enable S3 Versioning & Cross-Region Replication (CRR)

Lab Objective-

In this lab, you will:

- •Enable S3 Versioning
- •Configure Cross-Region Replication (CRR)
- •Validate replication between buckets

Prerequisites

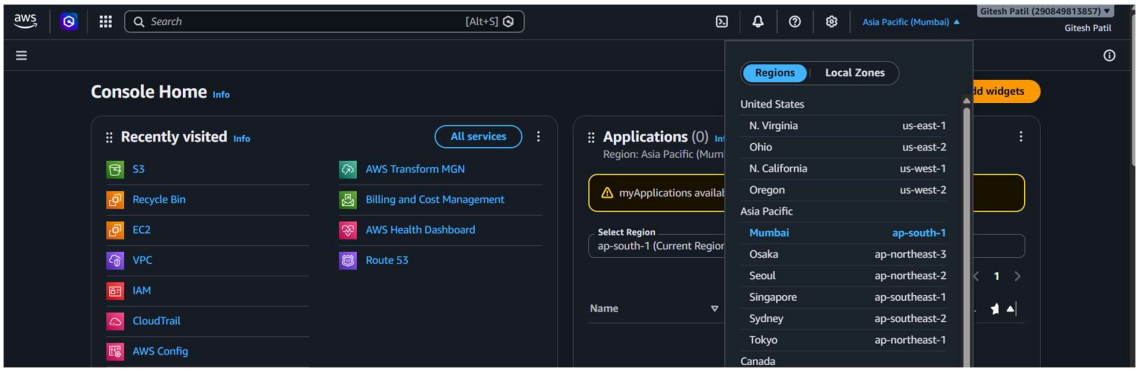
- • AWS account
- • Permissions for S3 & IAM
- • Two regions selected (e.g., ap-south-1 & us-east-1)

Lab Steps

Step 1: Log in & Select Source Region

1. Login to AWS Console

2. Select region (e.g., ap-south-1)



Step 2: Create Source S3 Bucket

1. Go to S3

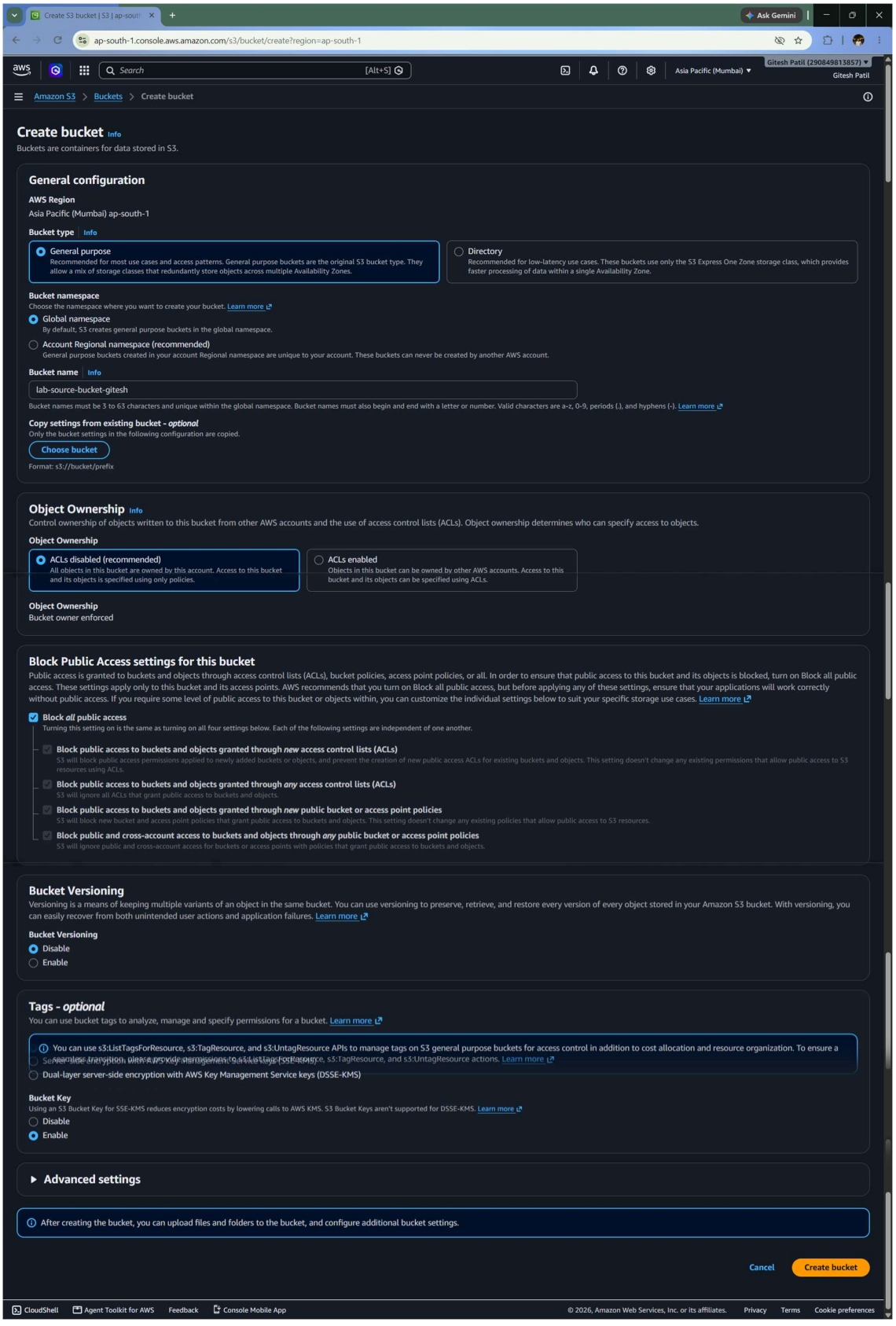
2. Click Create bucket

3. Enter:

- O Bucket name: lab-source-bucket-gitesh
- o Region: Source region

4.Disable public access (leave default)

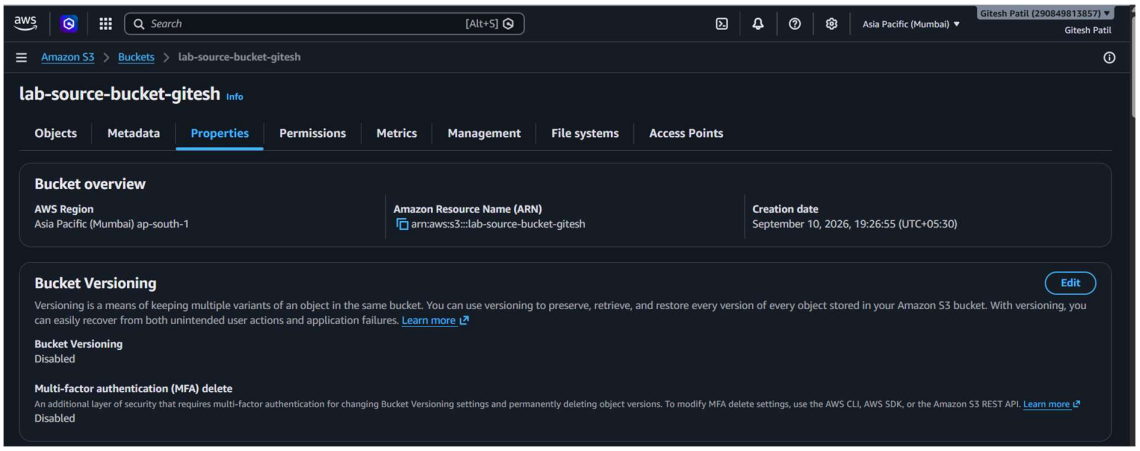
5.Click Create bucket


Create S3 bucket | S3 | ap-south
Ask Gemini
ap-south-1.console.aws.amazon.com/s3/bucket/create?region=ap-south-1
Search
[Alt+S]
Asia Pacific (Mumbai)
Gitesh Patil (290849813857)
Gitesh Patil
Amazon S3 > Buckets > Create bucket
Create bucket Info
Buckets are containers for data stored in S3.
General configuration
AWS Region
Asia Pacific (Mumbai) ap-south-1
Bucket type Info
General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
Directory
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.
Bucket namespace
Choose the namespace where you want to create your bucket. Learn more
Global namespace
By default, S3 creates general purpose buckets in the global namespace.
Account Regional namespace (recommended)
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.
Bucket name Info
lab-source-bucket-gitesh
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn more
Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix
Object Ownership Info
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.
Object Ownership
ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.
Object Ownership
Bucket owner enforced
Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more
Block all public access
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
Block public access to buckets and objects granted through new access control lists (ACLs)
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
Block public access to buckets and objects granted through any access control lists (ACLs)
S3 will ignore all ACLs that grant public access to buckets and objects.
Block public access to buckets and objects granted through new public bucket or access point policies
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
Block public and cross-account access to buckets and objects through any public bucket or access point policies
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.
Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more
Bucket Versioning
Disable
Enable
Tags - optional
You can use bucket tags to analyze, manage and specify permissions for a bucket. Learn more
You can use s3:ListTagsForResource, s3:TagResource, and s3:UntagResource APIs to manage tags on S3 general purpose buckets for access control in addition to cost allocation and resource organization. To ensure a seamless transition, please provide permissions to s3:ListTagsForResource, s3:TagResource, and s3:UntagResource actions. Learn more
Server-side encryption with AWS Key Management Service keys (SSE-KMS)
Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. Learn more
Disable
Enable
Advanced settings
After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.
Cancel
Create bucket
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

Step 3: Enable Versioning (SOURCE)

1. Open source bucket

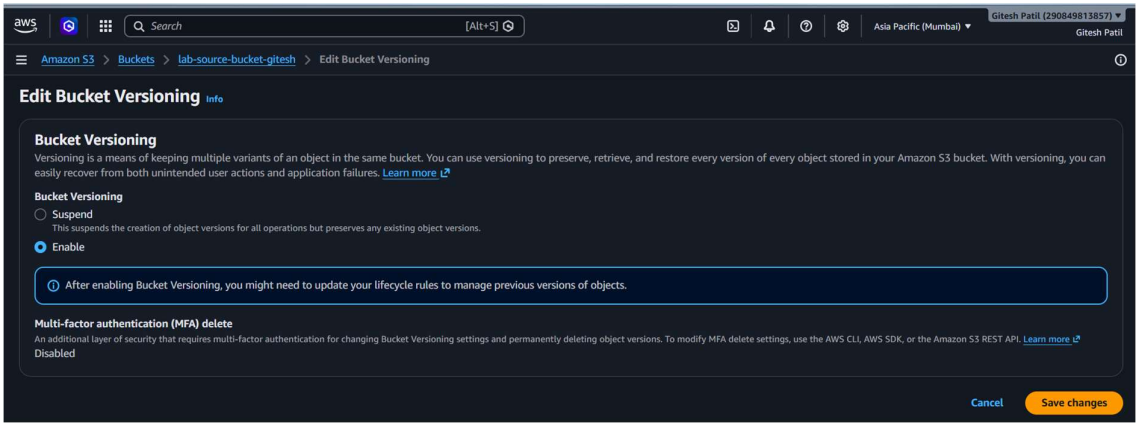
2. Go to Properties



3. Scroll to Bucket Versioning

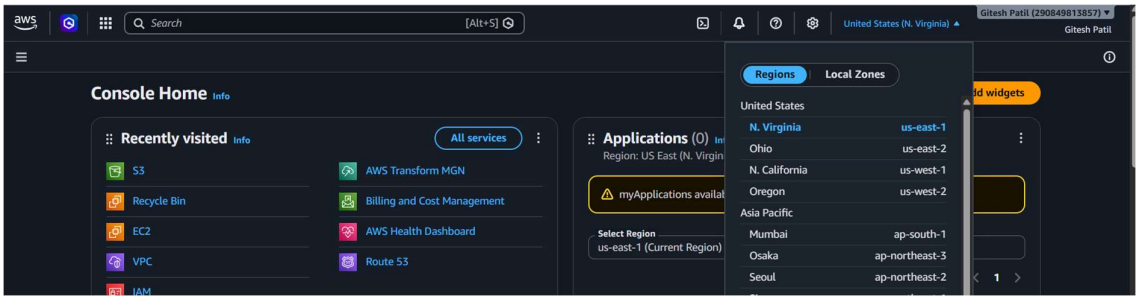
4. Click Edit

5. Enable Save



Step 4: Create Destination Bucket (Different Region)

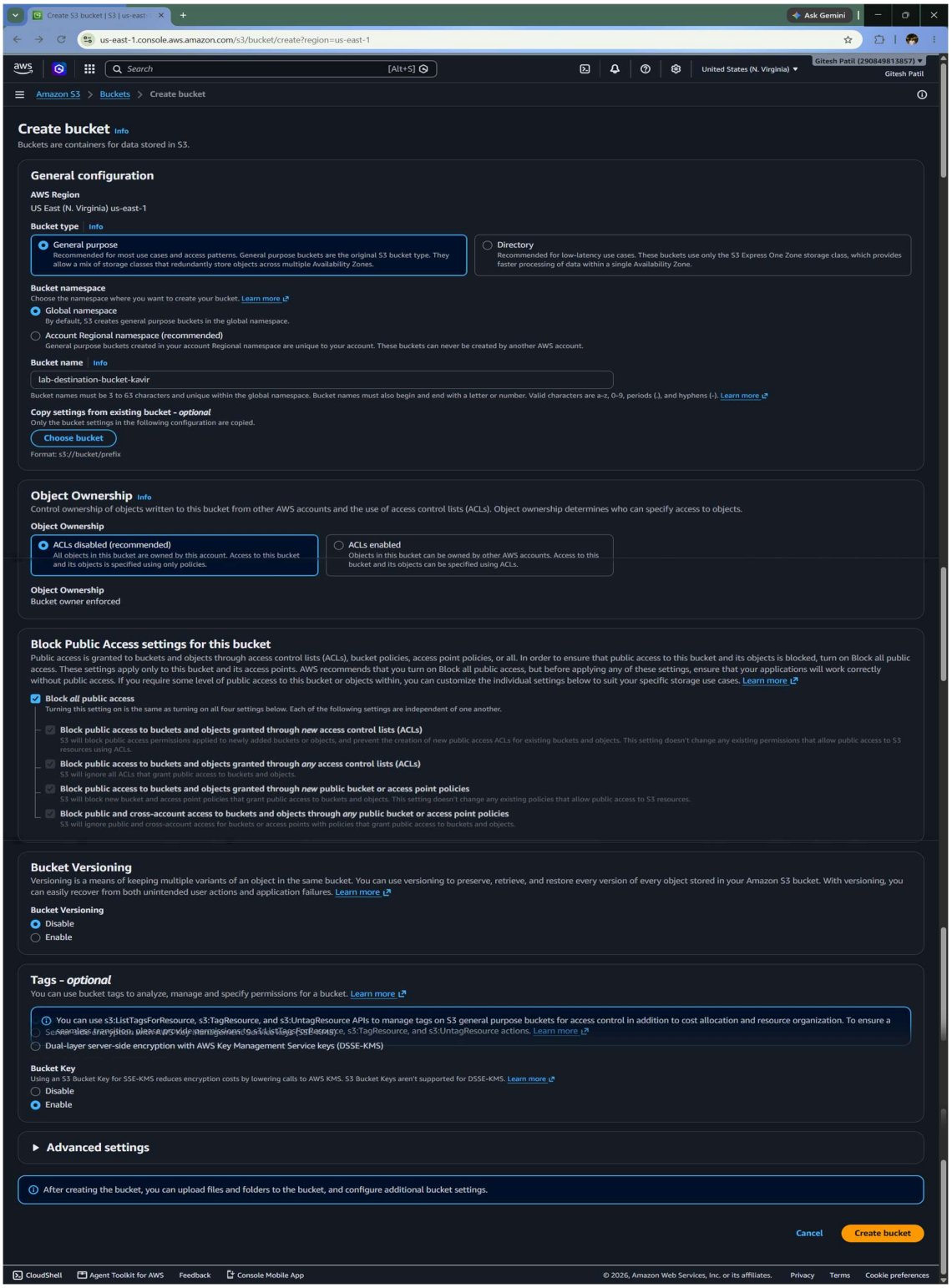
1. Switch region (e.g., us-east-1)

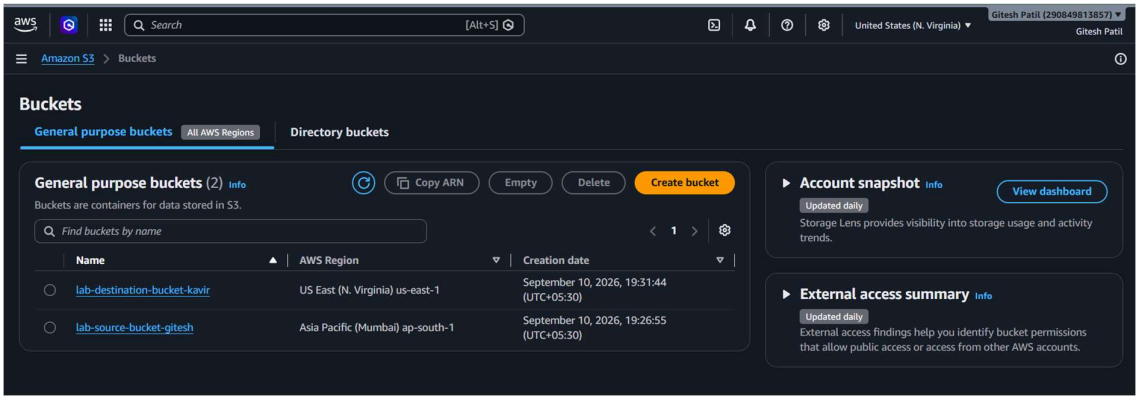


2. Click Create bucket

3. Name: lab-destination-bucket-kavir

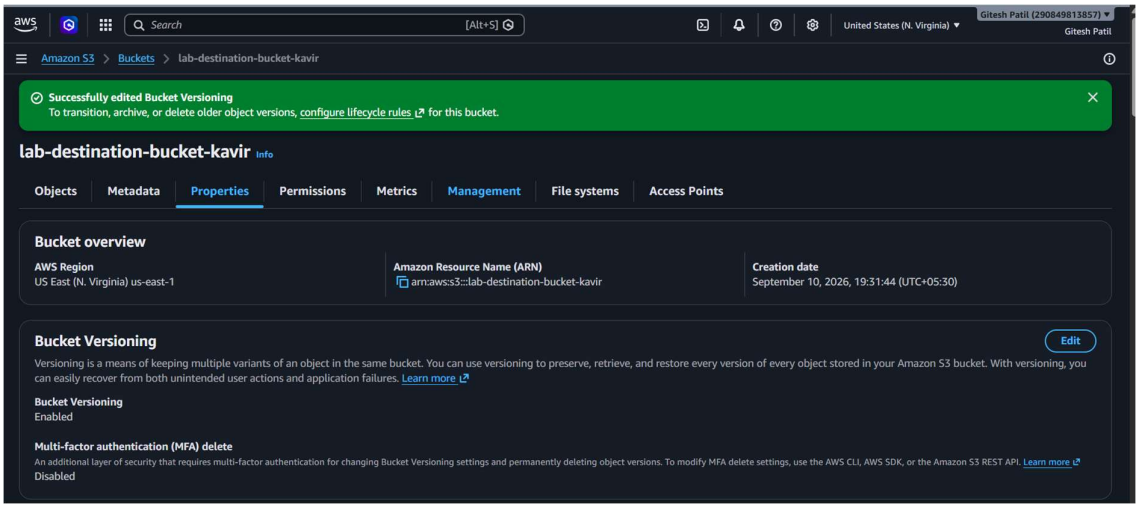
4. Create bucket





Step 5: Enable Versioning (DESTINATION)

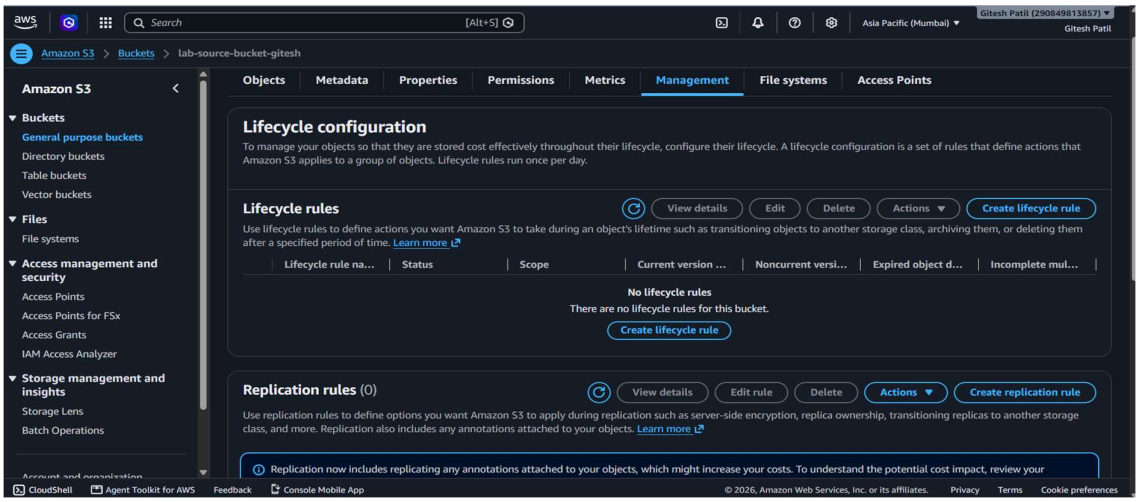
Repeat Step 3 for destination bucket.



Step 6: Configure Cross-Region Replication

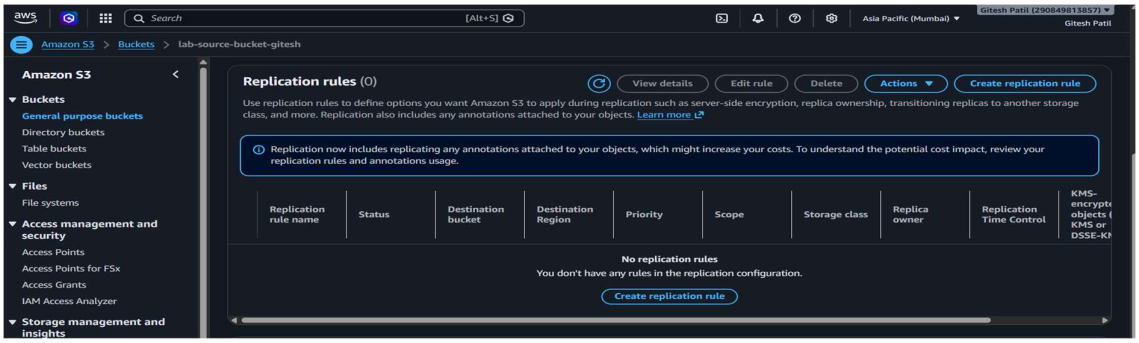
1. Go back to source bucket

2. Open Management tab



3. Click Replication rules

4. Click Create replication rule

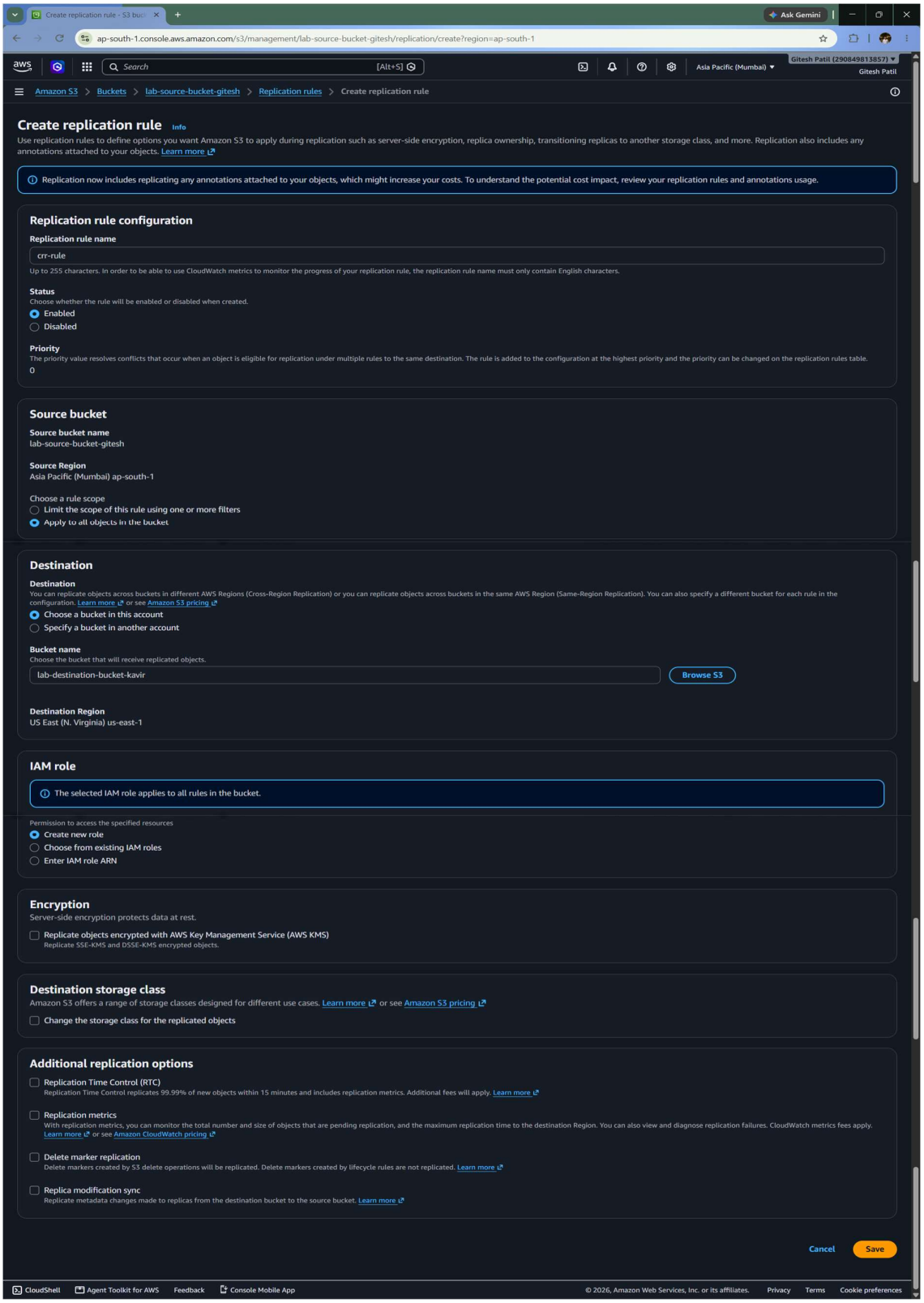


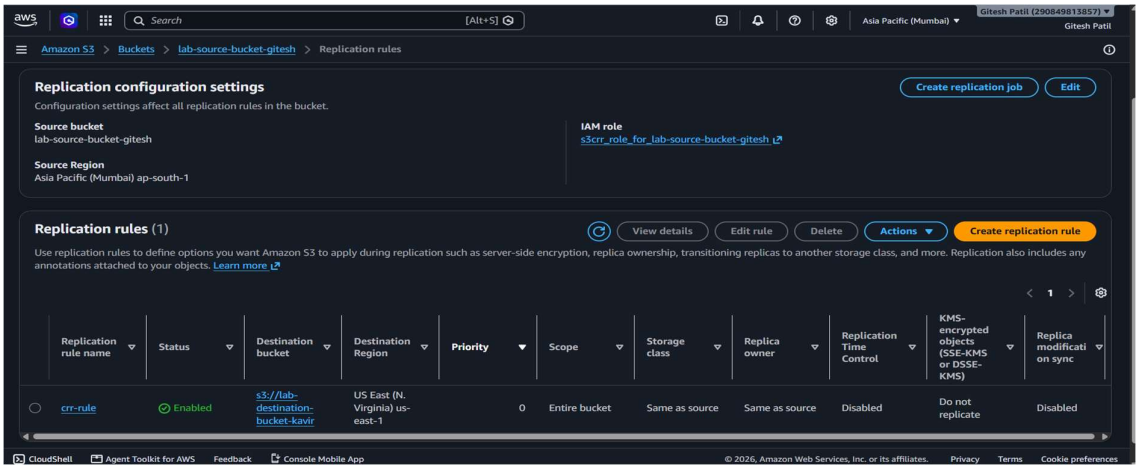
Step 7: Define Replication Rule

1. Rule name: crr-rule
2. Status: Enabled
3. Apply to: All objects

Step 8: Set Destination Bucket

1. Choose: Destination
2. Select Choose a bucket in this account
3. Click on Browse S3
4. Select destination bucket
5. Select Create new role
6. Click Save

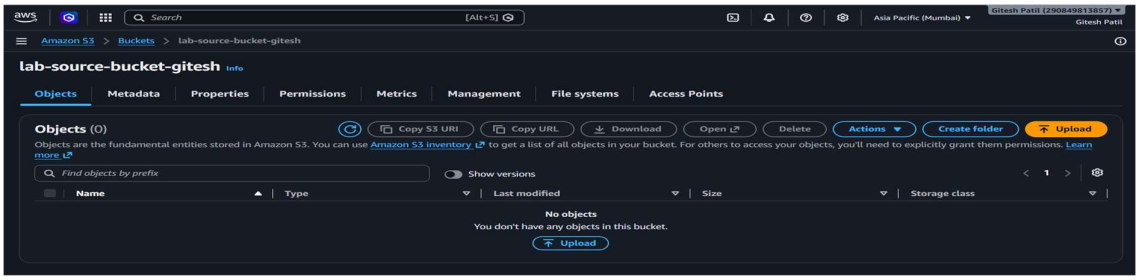

Create replication rule - S3 buc
Ask Gemini
ap-south-1.console.aws.amazon.com/s3/management/lab-source-bucket-gitesh/replication/create?region=ap-south-1
Search
[Alt+S]
Asia Pacific (Mumbai)
Gitesh Patil (290849813857)
Gitesh Patil
Amazon S3 > Buckets > lab-source-bucket-gitesh > Replication rules > Create replication rule
Create replication rule Info
Use replication rules to define options you want Amazon S3 to apply during replication such as server-side encryption, replica ownership, transitioning replicas to another storage class, and more. Replication also includes any annotations attached to your objects. Learn more
Replication now includes replicating any annotations attached to your objects, which might increase your costs. To understand the potential cost impact, review your replication rules and annotations usage.
Replication rule configuration
Replication rule name
crr-rule
Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain English characters.
Status
Choose whether the rule will be enabled or disabled when created.
Enabled
Disabled
Priority
The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be changed on the replication rules table.
0
Source bucket
Source bucket name
lab-source-bucket-gitesh
Source Region
Asia Pacific (Mumbai) ap-south-1
Choose a rule scope
Limit the scope of this rule using one or more filters
Apply to all objects in the bucket
Destination
Destination
You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the same AWS Region (Same-Region Replication). You can also specify a different bucket for each rule in the configuration. Learn more or see Amazon S3 pricing
Choose a bucket in this account
Specify a bucket in another account
Bucket name
Choose the bucket that will receive replicated objects.
lab-destination-bucket-kavir
Browse S3
Destination Region
US East (N. Virginia) us-east-1
IAM role
The selected IAM role applies to all rules in the bucket.
Permission to access the specified resources
Create new role
Choose from existing IAM roles
Enter IAM role ARN
Encryption
Server-side encryption protects data at rest.
Replicate objects encrypted with AWS Key Management Service (AWS KMS)
Replicate SSE-KMS and DSSE-KMS encrypted objects.
Destination storage class
Amazon S3 offers a range of storage classes designed for different use cases. Learn more or see Amazon S3 pricing
Change the storage class for the replicated objects
Additional replication options
Replication Time Control (RTC)
Replication Time Control replicates 99.99% of new objects within 15 minutes and includes replication metrics. Additional fees will apply. Learn more
Replication metrics
With replication metrics, you can monitor the total number and size of objects that are pending replication, and the maximum replication time to the destination Region. You can also view and diagnose replication failures. CloudWatch metrics fees apply. Learn more or see Amazon CloudWatch pricing
Delete marker replication
Delete markers created by S3 delete operations will be replicated. Delete markers created by lifecycle rules are not replicated. Learn more
Replica modification sync
Replicate metadata changes made to replicas from the destination bucket to the source bucket. Learn more
Cancel
Save
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



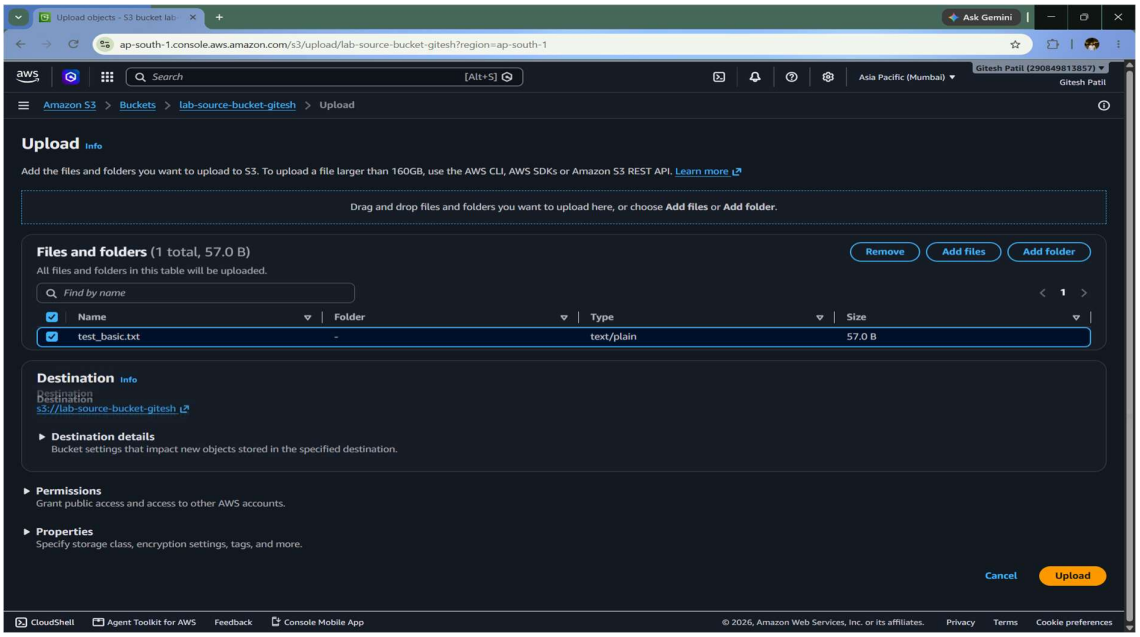
Step 11: Upload Test Object

1. Go to source bucket

2. Click Upload



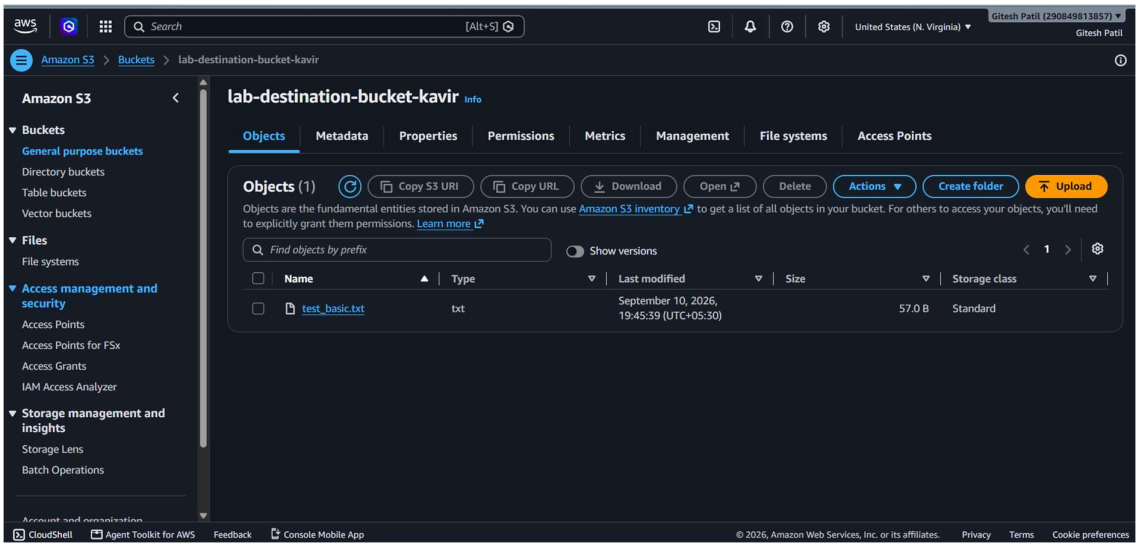
3. Upload any file (e.g., test_basic.txt)



4. Click on Upload

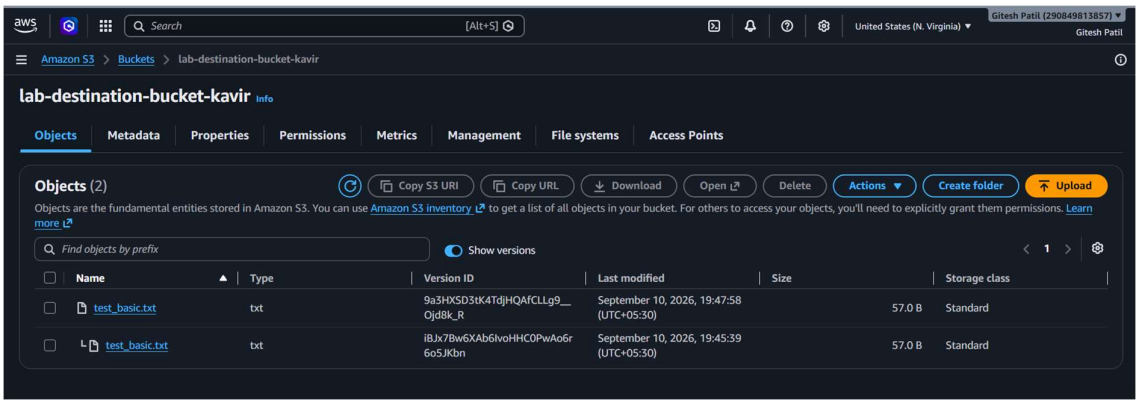
Step 12: Verify Replication

1. Switch to destination region
2. Open destination bucket
3. Check object exists



Step 13: Verify Versioning

1. Enable "Show versions"
2. Upload same file again
3. Check multiple versions exist



Step 14: Cleanup

1. Delete objects in both buckets
2. Delete buckets